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1 June 1984
SUPERSEDING
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26 Februarv 1974

MILITARY SPECIFICATION

TOOL SETS, SHOP SETS AND KITS

MAINTENANCE, MODIFICATION AND TOOL

PACKAGING OF

This specification is approved by use for all Department and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers the requirements for methods of preservation, packing, and marking of components for both common and special tool sets, shop sets, and kits (hardware, installation, maintenance, modification, and tool) for handling, shipment, and storage. (This specification does not include tool sets or tool kits that are furnished in plastic, cloth, or leather cases, or relatively small wooden or metal carrying cases (see 6.3).

1.2 Classification. Tool sets, kits, and shop sets shall be of the following types, as specified (see 6.2):

Type I - Kits and sets shall be the complete assemblage of tools, packaged in a single carrying case.

Type II - Kits and sets shall be the complete assemblage of tools, equipment, and compressed gases requiring two or more shipping containers.

Type III - Kits and sets shall be the complete assemblage of tools and equipment requiring a single multi-packed box (less carrying case).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-LEE-S, Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC PACK

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2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-81	- Tags, Shipping and Stock
PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-B-1055	- Barrier Material, Waterproofed, Flexible
PPP-C-795	- Cushioning Material, Flexible, Cellular, Plastic Film for Packaging Application
PPP-C-843	- Cushioning Material, Cellulosic
PPP-C-1120	- Cushioning Material, Uncompressed Bound Fiber for Packaging
PPP-C-1797	- Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam
PPP-H-1581	- Hardware (Fasteners and Related Items) Packaging and Packing For Shipment and Storage of
PPP-P-40	- Packaging and Packing of Hand Tools
PPP-P-115	- Pads, Paper Macerated
PPP-P-291	- Paperboard, Wrapping and Cushioning
PPP-T-42	- Tape, Packaging/Masking, Paper
PPP-T-60	- Tape, Packaging Waterproof
PPP-T-1150	- Tools and Tool Accessories for Power Driven, Metal and Woodworking, Machinery; Packaging and Packing of

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	- Preservation, Methods of
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-F-3296	- Forges, Furnaces and Ovens (Exclusive of Space Heating and Cooking), Packaging of
MIL-P-3420	- Packaging Materials, Volatile Corrosion Inhibitor Treated, Opaque
MIL-C-3600	- Compressors, Air and Gas (Except Oxygen) Packaging of
MIL-C-3774	- Crates, Wood; Open 12,000- and 16,000-Pound Capacity
MIL-B-3865	- Blocks, Rope, Tackle: Packaging of
MIL-C-5501	- Cap and Plug, Protective, Dust and Moisture Seal General Specification for

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MIL-I-8574 - Inhibitors, Corrosion, Volatile, Utilization of
MIL-T-14527 - Pole Line Hardware: Preparation For Delivery of
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of
MIL-M-18058 - Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation For Delivery of
MIL-G-28554 - Generator Sets, Mobil Electric Power and Supplemental Equipment, Packaging of
MIL-B-43666 - Boxes, Shipping Consolidation
MIL-V-45554 - Vulcanizing Equipment, Including Related Items, Tire and Tube, Rebuild and Repair; Preparation For Delivery of
MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of
MIL-B-45977 - Bins, Cabinets, Shelving, and Worktables, Packaging of
MIL-C-52950 - Crates, Wood, Open and Covered

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

TM 38-250/AFR 71-4/NAVSUPPUB505/MCO P 4030.19D/DLAR 4145.3 - Preparation of Hazardous Materials for Military Air Shipment

DoD 4145.19-R-1 - Storage and Materials Handling

DoD 4500.32-R - Military Standard Transportation and Movement Procedures

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

TARIFF BOE-6000-C - Bureau of Explosives Tariffs (CFR 49, Parts 106-177)

(Application for copies should be addressed to Thomas A. Phemister Agent, Association of American Railroads, Bureau of Explosives, 1920 "L" Street N.W., Washington, D.C. 20036.)

TARIFF 6-D - Commercial Air Shipments

(Application for copies should be addressed to Airline Tariff Publishing Company, Dallas International Airport, P.O. Box 17415, Washington, D.C. 20041.)

INTER-GOVERNMENTAL MARITIME CONSULTATIVE ORGANIZATION (IMCO)

(Application for copies should be addressed to Whitstable Litho, Limited, Whitstable Kent, Great Britain)

INTERNATIONAL AIR TRANSPORT ASSOCIATION (IATA)

(Application for copies should be addressed to Publications Agent, 2000 Pell Street, Montreal, Quebec. Canada H3A 2R4)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish one complete pack, including up to date packing list, for first article inspection. The first article may be either a preproduction pack or an initial production pack which conforms to the requirements of this specification. In either case, the approved first article and the production packs shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Component grouping. For the purpose of this specification, component parts and assemblies have been segregated into groups according to the composite characteristics as specified in Table I in order to determine the applicable method of protection required.

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Table I. Composite characteristics component grouping

Group	Composition	Description
1	Metallic, Ferrous	Items in this group are comprised wholly of metals and alloys containing an appreciable quantity of iron, including the many types of steel. Surfaces may be bare, partially plated, coated, or painted. Items of this nature include, but are not limited to, hoists, tow chains, bolt cutters, workbenches, anvils, tow bars, auto jacks, vises, and common hand tools.
2	Metallic, including Non-ferrous and Painted or Plated Ferrous	Items in this group are comprised wholly of metallic, nonferrous metals and items which are completely painted or plated or inherently corrosion resistant. Items of this nature include, but are not limited to, nonsparking hand tools such as brass hammers, copper tipped hammers, and aluminum tipped hammers. Also, folding tables, crowbars, and tool cabinets.
3	Metallic, and Nonmetallic Composites	Items in this group are comprised of two or more materials, one of which being metallic and the remaining materials nonmetallic. Items of this nature include, but are not limited to, tire gages, electric soldering irons, files and reamers with wooden handles, electric lanterns, hand saws, claw hammers, axes, and wire brushes.
4	Nonmetallic	Items in this group are comprised of nonmetallic materials regardless of whether the item is comprised wholly of one nonmetallic material or a combination of nonmetallic materials. Items of this nature include, but are not limited to, technical data, leather and rubber aprons, tool bags, goggles, safety gloves, and abrasive products.
5	Electrical and Electronic Equipment	Items in this group include electronic equipment. Items of this nature include, but are not limited to, test sets, electric hand tools, motors, and complete electronic systems.

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TABLE I. Composite characteristics component grouping-Continued.

Group	Composition	Description
6	Pneumatic Hand Tools	Items in this group are tools that are air power driven and include, but are not limited to, hammers, sanders, polishers, bucking tools, riveters, and impact wrenches.
7	Complex Items	Items included in this group are individually preserved, packaged, and packed because of their construction, configuration, size, and weight. Items of this nature include, but are not limited to, metal and woodworking machines, compressors, generators, welders, steam cleaners, and sewing machines.
8	Compressed Gases	Items in this group are limited to compressed gases.

3.3 Materials. Packaging materials shall be as specified herein. Material shall be new and free of defects.

3.4 Disassembly and matchmarking.

3.4.1 Disassembly. Disassembly shall be limited to the extent necessary to accomplish necessary reduction in cube, protect parts vulnerable to damage, ensure thorough cleaning and drying, and preservative application. A part of an assembly shall not be removed unless it can be readily assembled in the field without special tools or skilled personnel.

3.4.2 Matchmarking. When necessary to facilitate reassembly, removed parts and assemblies shall be matchmarked. Matchmarking information shall be annotated on shipping tags conforming to UU-T-81, type B, class 2 and be attached to mating parts.

3.5 Processing conditions. Preservation shall be accomplished within buildings which have facilities to prevent the accumulation of dust and moisture on the equipment being processed. Equipment brought into processing areas from the outside shall be allowed sufficient time to reach ambient temperatures before processing. Cleaning, drying, application of preservatives, and sealing of barriers shall progress in an uninterrupted series of operations. When periods of interruption are necessary temporary protection shall be provided to partially processed items, as required, to ensure against contamination.

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3.6 Cleaning and drying. Thorough cleaning and drying shall be accomplished without damage to any part. Cleaning of all exposed surfaces of parts and assemblies shall be accomplished through one or a combination of processes referenced in MIL-P-116. Drying shall be accomplished through one or more drying procedures referenced therein. Special cleaning processes and drying procedures may be employed upon approval of the procuring activity concerned.

3.7 Preservatives. Preservatives specified herein, identified by symbol, e.g., type P-1, shall conform to the applicable specifications listed in MIL-P-116.

3.8 Stowage compartments. When stowage compartments are provided in the carrying case, the kit components shall be stowed within the designated compartments to the extent practicable. Overflow of packaged items shall be placed in tiers inside the carrying case except when the carrying case will not adequately house the complete packaged kit, the overflow of parts will be placed inside the unit container (see 3.10.1.1.5) along with the carrying case.

3.9 Unit protection. The methods of preservation specified herein shall be in accordance with MIL-P-116.

3.10 Preservation. Preservation shall be level A, level B, or Commercial as specified (see 6.2).

3.10.1 Level A. Components of tool sets and kits shall be preserved and unit packaged in accordance with methods specified herein. Tool and equipment items shall be unit packaged in quantities specified in the applicable packaging data document. Alternatively, diminutive (small) type items, bearing the same National Stock Number, may be unit packaged in multiple quantities with the following limitations:

a. Maximum size of 24 cubic inches for individual items.
b. Maximum of 100 pieces per unit package.
c. Maximum net load of 10 pounds.

The unit method of protection specified in the packaging data document shall in no case be down graded when packaging items in multiple quantities. However, the submethod may be changed to better accommodate the load, e.g., submethod IC-1 to IC-2 and IA-8 to IA-15.

3.10.1.1 Type I. Type I kits or tool sets shall be preserved and packaged in accordance with the applicable methods listed below. The selection of methods and application of materials shall be in accordance with the criteria and guidelines of MIL-P-116 and MIL-STD-794.

3.10.1.1.1 Method I utilizing Volative Corrosion Inhibitor (VCI) (P-18). The preferred method of packaging type I kits or tool sets is method I utilizing VCI treated materials, in place of a contact preservative. The VCI material shall conform to MIL-P-3420, type I and be used in accordance with "Limitation on use" in MIL-I-8574. Kits or tool sets packaged by this method will be mainly comprised of items in groups 1 and 3 or a combination thereof. (This method shall not be used for items in group 5.)

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3.10.1.1.2 Method I utilizing contact preservative. Component parts that cannot be packaged with VCI treated materials, because of their composition or surface finish (see MIL-T-8574), shall be packaged in accordance with method I, utilizing a contact preservative. Component parts that are useable without removing the preservative shall be coated with type P-1 or P-19 preservative. Thoroughly dried parts need not be wrapped. Unless otherwise specified in commodity or packaging specifications, critical ferrous metal surfaces shall be preserved with type P-2 or type P-6 preservative. Surfaces preserved with type P-2 or type P-6 preservative shall be wrapped with barrier material conforming to MIL-B-121, grade A, class 2. Barrier material shall be secured with tape conforming to PPP-T-42. Tools preserved by this method will be mainly comprised of items in groups 1, 3, and 6.

3.10.1.1.3 Method II. Unless otherwise specified in commodity or packaging specifications, kits or tool sets comprised of items in group 5 shall be packaged in accordance with the applicable submethod of method II. The carrying case may serve as the inner container when submethods IIb and IIe are employed (see 3.9).

3.10.1.1.4 Method III. Method III shall be utilized in packaging type I kits or tool sets comprised of items in group 2.

3.10.1.1.5 Unit container. Type I kits and sets shall be individually packaged in a snug-fitting fiberboard box conforming to PPP-B-636, class weather-resistant. The box closure shall be in accordance with the appendix to the box specification.

3.10.1.1.6 Cushioning. Cushioning material shall be employed inside the carrying case to restrict movement and protect parts that are vulnerable to damage. Material shall conform to any of the following: PPP-C-843, PPP-C-1797, PPP-C-795, PPP-C-1120, PPP-P-115, or PPP-P-291. All fragile and delicate items, or items having cutting edges, shall be isolated to prevent impact with other components.

3.10.1.2 Type II and III. Components of types II and III tool sets and shop sets, categorized by composition characteristics of the various groups of Table I, shall be preserved and packaging as follows:

Group 1. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of MIL-T-14527, MIL-W-45562, MIL-B-3865, PPP-H-1581, MIL-V-45554, PPP-T-1150, or PPP-P-40, as applicable. In the event the packaging specification does not cover the item, package in accordance with method I of MIL-P-116.

Group 2. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of PPP-H-1581, MIL-T-14527, MIL-B-45977, MIL-W-45562, or PPP-P-40, as applicable. In the event the packaging specification does not cover the item, package in accordance with method III of MIL-P-116.

Group 3. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of MIL-T-14527, PPP-P-40, MIL-B-45977, MIL-B-3865, or MIL-W-45562, as applicable. In the event the packaging specification does not cover the item, package delicate and fragile items in accordance with method IC and rugged-type items method I of MIL-P-116.

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Group 4. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of PPP-P-40. In the event the packaging specification does not cover the item, package in accordance with method IC of MIL-P-116.

Group 5. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of MIL-E-17555, MIL-E-16298, PPP-P-40, or the commodtiy specification, as applicable. In the event the packaging specification does not cover the item, package in accordance with method II of MIL-P-116.

Group 6. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of PPP-P-40.

Group 7. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of MIL-M-18058, MIL-F-3296, MIL-C-3600, MIL-E-16298, MIL-G-28554, or the applicable commodity specification.

Group 8. Items having composition characteristics of this group shall be packaged in accordance with Tariff BO E-6000-C (CFR 49, Parts 106-177) for geographical North America shipments; Inter-Governmental Maritime Consultative Organization (IMCO) for international vessel; International Air Transport Association (IATA) for international air; Tariff 6-D for commercial domestic shipments, and AFR 71-4/TM 38-250/NAVSUPPUB 505/MCO P 4030.19D/ DLAR 4145.3 for military air shipments.

Transportation through the Defense Transportation System is covered in DoD 4500.32-R. In addition to the above, packaging requirement cyclinders must have their valves protected by one of the following methods:

a. By a securely attached metal cap of sufficient strength to protect the valve from injury during transit.
b. By boxing or crating the cylinder or sphere so as to give proper protection to the value.
c. By a recessed valve or otherwise by a protective valve so that it will be protected during transit.

3.10.1.3 Alternate method of preservation. Ferrous metal items which are received or stored at a DoD installation, that are coated with a preservative compound, other than that specified by the applicable packaging data document, or coated with laquer or varnish, will be acceptable for level A protection, unless evidence of deterioration or corrosion is present.

3.10.1.4 Alternate method of unit protection. For set components which are scheduled to be unit packaged method I or method III, that utilize a fiberboard, wood, or paperboard box as the unit container, unit containers may be omitted, if the consolidation container is water-resistant (see 3.11.1.4), or when the item is packed as specified in 3.11.1.3. If unit container is not omitted the use of Domestic Grade Fiberboard is acceptable for Level A protection when packed within Level A consolidation container (see 3.11.1.4). The water-resistant protection will come from the exterior consolidation container in lieu of the unit container. Critical or

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painted surfaces may require additional protection against abrasion or physical damage. These surfaces shall be protected by wrapping or covering with cushioning material specified in 3.10.1.1.6; package the item in a multi-pack container, or cover the surface with tape or cap-plugs conforming to PPP-T-60, type optional or MIL-C-5501, respectively.

3.10.2 Intermediate packaging. Type I kits or tool sets weighing ten pounds or less, bearing the same stock number, shall be intermediate packaged in a close-fitting box conforming to PPP-B-636, type CF, class weather- resistant. The gross weight of the intermediate package shall not exceed 35 pounds nor displace more than 1.5 cubic feet. Box closure shall be in accordance with the appendix to the box specification.

3.10.3 Commercial preservation. Commercial preservation of tool sets, shop sets, and kits shall conform to the requirements of ASTM D3951-82.

3.10.4 Army procurements.

3.10.4.1 Level A and level B unit packing. Unit packs shall be as specified except that all unit packs in fiberboard containers shall be class weather-resistant and for level B unit packs, the methods of preservation shall be substituted as follows:

Level A	Level B (substitute method)	Remarks
IA-8	IC-1	
IA-14	IC-2	Omit outer container and wrap with Kraft paper.
II	IA	Omit desiccant.
III		No change.

3.10.4.2 Level A and level B intermediate packs. All intermediate containers shall either be weather (or water) resistant or overwrapped with waterproof barrier materials. Containers conforming to PPP-B-566 or PPP-B-676 shall be variety 2, or be overwrapped with waterproof barrier materials. Intermediate containers shall be uniform in size, shape, and quantities; shall be of minimum tare and cube; and shall contain multiples of 5 unit packs not to exceed 100 unit packs. Intermediate containers shall not exceed a maximum of 40 pounds net weight and a maximum of 1.5 cubic feet, with at least each of two dimensions not exceeding 16 inches. Intermediate containers shall not be required when the total quantity being shipped will result in only one intermediate pack per shipping container.

3.11 Packing. Packing shall be level A or B, or Commercial, as specified (see 6.2).

3.11.1 Level A.

3.11.1.1 Type I. Kits or tool sets shall be packaged in a box conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-585, class 3, style 2. Box closure and strapping requirements shall be in accordance with the applicable box specification and the appendix thereto. Gross weight of the shipping container shall be governed by the weight limitations of the applicable container specification.

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3.11.1.2 Type II. Components of kits and shop sets shall be packed in two or more exterior shipping containers. Items displacing three cubic feet or more shall be packed as specified in 3.11.1.2.1. Items displacing less than three cubic feet shall be packed as specified in 3.11.1.3. Compressed gas cylinders having a protective cap over the valve assembly may be shipped uncrated. Box closure and strapping requirements shall be in accordance with the applicable container specifications and the appendix thereto. In the event an open crate is utilized as the shipping container, a waterproof shroud shall be placed over the item. Anchoring, shrouding, blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. (NOTE: Items packaged in accordance with method II requirements of MIL-P-116 shall not be packed in open crates.)

3.11.1.2.1 Individual pack. All items displacing three cubic feet or more shall be individually packed. Shipping containers shall conform to PPP-B-585, class 3, styles 2 or 3; PPP-B-601, overseas type; PPP-B-621, class 2; MIL-C-52950; MIL-C-3774. Consideration will be given the following factors when selecting a suitable shipping container:

a. Adequate protection for the commodity against physical damage.
b. The most economical container.
c. Minimum cube and tare.

3.11.1.3 Type III (consolidated pack). Packaged items displacing less than three cubic feet shall be consolidated and packed in a box conforming to PPP-B-621, class 2; PPP-B-601, overseas type; or MIL-B-43666, type I or type II. Selected items shall be placed in the shipping container in accordance with their fragility characteristics. Noncritical, rugged-type items shall be placed in the container first and critical, fragile-type items shall be positioned at the top of container. Blocking, bracing, and cushioning of the items shall be as specified in 3.11.1.2.

3.11.1.4 Consolidation container. Unless otherwise specified, compliance to individual packaging data sheets is waived when the tool set shipping containers conform to Style A of PPP-B-601 (overseas type), or class 2, Style 2 or 4 of PPP-B-621 except the tops shall be 1/2" thick, plywood sheeting bonded with exterior type glue, or PPP-B-1055, waterproofed barrier material, covered with nominal 1" lumber. Sides and ends of PPP-B-621 boxes shall be lined with PPP-B-1055, waterproofed barrier material. The use of up to date packaging materials, such as cohesive VCI impregnated polyethylene, as an alternate to method IC, and cohesive coated cushioning materials, are encouraged. Items received, in accordance with MIL-STD-1188, from depots or procurement already packaged to an adequate degree of protection for line item stockage and issue need not be reprocessed for shipment in accordance with individual packaging data sheets.

3.11.1.5 Closure. Box closure, skids and strapping for shipping containers shall be in accordance with the appendix to the applicable box specification.

3.11.2 Level B. Types I, II, and III kits, tool sets, and shop sets shall be packed as specified for level A (see 3.11.1), except boxes utilized in shipping shall conform to PPP-B-601, domestic type, styles A, B, or J; PPP-B-621, class 1; or PPP-B-640, class 2 in lieu of those specified for level A. Closurers see (3.11.1.5).

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3.11.3 Commercial packing. Commercial packing of tool sets, shop sets, and kits shall conform to the requirements of ASTM 03951-82.

3.11.4 Unitized load. When specified (see 6.2), a quantity of kits or sets shall be unitized in accordance with MIL-STD-147. A fiberboard cap shall be employed over the load as specified therein.

3.12 Performance.

3.12.1 Rough handling. When specified (see 6.2), the completed pack shall provide protection from rough handling sufficient to prevent damage to the packaged item(s) when the pack is subjected to the rough handling tests of MIL-P-116. Following the tests, the packaged item(s) shall show no evidence of damage, and the exterior container shall not reveal any failure caused by improper application or design of interior cushioning, blocking, or bracing. All barriers, wrappings, interior containers, blocking, bracing and anchoring shall be intact and capable of providing the intended protection.

3.12.2 Exposure. When specified (see 6.2) the completed pack shall provide protection from exposure to the elements sufficient to prevent damage to the packaged item(s) when the pack is subjected to the cyclic exposure test of MIL-P-116. Following the test, the tested package, when opened, shall show no evidence of moisture nor shall the packaged item show corrosion.

3.13 Marking.

3.13.1 Military markings. When military levels of protection are specified, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and any special markings required by the contract or purchase order.

3.13.1.1 Cylinders. Painted surfaces of compressed gas cylinders shall not be defaced with markings except for the application of hazardous precautionary labels. Cylinders scheduled to be shipped uncrated shall be marked by the application of shipping tags. Identification markings and set markings shall be applied to tags conforming to UU-T-81, type A or B, class 2. Address markings shall be annotated on Military Shipping Tag DD Form 1387-1. To preserve the cylinders' identity and shipping address, two tags reflecting like information shall be utilized. When practicable, tags shall be attached to the protective cap of the cylinder with a metal wire tie. Wire shall be not less than 23 gage (0.0258-inch diameter). When cylinders are crated for shipment, identification markings shall be applied as specified above.

3.13.2 Commercial marking. When commercial preservation and packing is specified, marking shall be in accordance with ASTM D3951-82. Special markings required will be as specified in the contract or purchase order.

3.13.3 Packing list. When level A, B, or commercial preservation is specified, an up to date packing list shall accompany each tool set, kit, or shop set. The packing list shall be prepared and utilized as specified in MIL-STD-129.

3.14 Storage. Items shall be stored in accordance with DoD 4145.19-R-1, Storage and Materials Handling.

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3.15 Workmanship. Workmanship shall be such that when the proper procedure is followed, materiel and equipment being processed will receive the maximum protection against corrosion, deterioration, and damage during handling, shipment, and storage.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements or purchase order as specified herein. Except as otherwise specified in the contract, or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such specifications are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. When first article inspection is required by the contract or purchase order (see 3.1), the contractor shall subject the first article pack to the applicable first article inspection and tests specified in MIL-P-116 and the examinations in the applicable container specification and Appendix B of MIL-STD-1186 to determine conformance to the requirements of this specification. When specified (see 6.2), first article packages shall be subjected to the rough handling and cyclic exposure test(s) (one or both) as specified in the first article inspection requirements of MIL-P-116. Failure of the first article to pass any of the inspections or tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to production packs offered for acceptance under the contract or purchase order.

4.2.2.1 Level A and B preservation and packing. Quality conformance inspection shall consist of the applicable quality conformance inspections and tests specified in MIL-P-116 and the examinations in the applicable container specification and Appendix B of MIL-STD-1186 to determine conformance to the requirements of this specification. When specified (see 6.2), the packs shall be subjected to the rough handling and cyclic exposure test(s) (one or both) as specified in the quality conformance inspection requirements of MIL-P-116.

4.2.2.2 Commercial preservation and packing. Commercial preservation and packaging shall be examined to determine conformance with the requirements of ASTM D3951-82.

4.3 Sampling.

4.3.1 Level A and B preservation and packing. Sampling for quality conformance inspection shall be conducted in accordance with MIL-STD-105 at the inspection level and acceptable quality level (AQL) specified in MIL-P-116 and Appendix B of MIL-STD-1186, respectively. Sampling for examination of the container shall be at level II and an AQL of 4.0. When specified (see 6.2), sampling for the rough handling and cyclic test(s) (one or both) shall be at inspection level S-2 with an AQL of 4.0.

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4.3.2 Commercial preservation and packing. Sampling for quality conformance inspection shall be conducted in accordance with MIL-STD-105 at inspection level S-4 with an AQL of 4.0.

4.3.3 Disposition of samples. All samples used for inspection and testing shall be reprocessed as necessary. They may, after reprocessing in accordance with the original method of preservation, be considered a part of the original lot. When the packaged item may have been damaged as a result of testing, the item shall be inspected as necessary to determine its acceptability.

5 PACKAGING

5.1 This section is not applicable to this specification.

6 NOTES

6.1 Intended use. It is intended that this specification be used for reference in section 5 of related commodity specifications and direct reference in procurement documents. The preservation, packaging, and marking requirements specified herein are intended to ensure tools and equipment against corrosion, deterioration, and damage, and to provide for proper identification during shipment and storage.

6.2 Ordering data.

a. Title, number, and date of specification.
b. Type I, II, or III tool sets, kits, or shop sets, and equipment (see 1.2).
c. First article, when required (see 3.1).
d. Selection of applicable degree of protection A, B, or Commercial (see 3.10 and 3.11).
e. Unitized load, when required (see 3.11.4).
f. First article inspection, if different (see 4.2.1).
g. Quality conformance inspection, if different (see 4.2.2).
h. Rough handling test, when required (see 3.12.1, 4.2.1).
i. Cyclic exposure test, when required (see 3.12.2, 4.2.1).
j. Level B preservation requirements shall be selected only under known favorable storage and shipping requirements.

6.3 Tool sets and kits (small). Tool sets and kits furnished in plastic, cloth, or leather carrying cases, or relatively small wooded or metal boxes, shall be packaged and packed in accordance with PPP-P-40.

6.4 Unit protection. When the environment within tool set consolidation boxes (see 3.11.1.4) provides water-resistant protection to the items packed therein, the equivalent protection cited on individual packaging data sheets becomes excessive. Individual change request procedures to reduce packaging requirements are ineffective due to the universal magnitude. Certain items may still require Level A unit protection due to critical, sensitive or crushable characteristics.

6.5 Disposability methods. Environmental Pollution Preventative Measures are contained in the materials' specifications indicated herein. Refer to the materials' specifications (or preparing activity) for recommended disposability methods.

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